

Software Engineering

Lab 1: Requirements Engineering & UML Use-Case Modelling

System / Project: Community Solar Credit Allocation Manager

Domain: Sustainability & Green Tech

Target Actors : Prosumer Resident, Co-op Manager

Elicit & Draft Requirements

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale	Comments
FR-001	Functional	The system shall allow a Prosumer to register the amount of solar energy generated.	High	When valid solar generation data is entered, the system records the generation amount and displays the updated value.	Required to determine available solar energy for credit allocation.	
FR-002	Functional	The system shall allow a Resident to view their allocated solar credit balance.	High	When the Resident opens the credit balance, the system displays the current allocated credit amount.	Allows residents to track their available solar credits.	
FR-003	Functional	The system shall allow the Co-op Manager to allocate available solar credits to eligible Residents.	High	When a valid allocation is submitted, the system updates the selected Resident's credit balance with the allocated amount.	Core function of the credit allocation system.	
FR-004	Functional	The system shall allow a Resident to view their solar credit allocation history.	Medium	When the Resident requests allocation history, the system displays previous allocations with their dates and amounts.	Provides transparency and tracking of previous allocations.	
FR-005	Functional	The system shall notify a Resident when their solar credit	Medium	After a successful allocation, the affected Resident receives an	Keeps residents informed about	

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		allocation is updated.		allocation notification.	changes to their credits.	
NFR-001	Nonfunc.	The system shall display a Resident's current solar credit balance within 3 seconds of the request.	High	The credit balance is displayed in less than 3 seconds for at least 95% of tested requests.	Ensures a responsive user experience.	
NFR-002	Nonfunc.	The system shall restrict credit allocation management to authorized Co-op Managers.	High	An unauthorized user attempting to access credit allocation management is denied access.	Prevents unauthorized changes to credit allocations.	

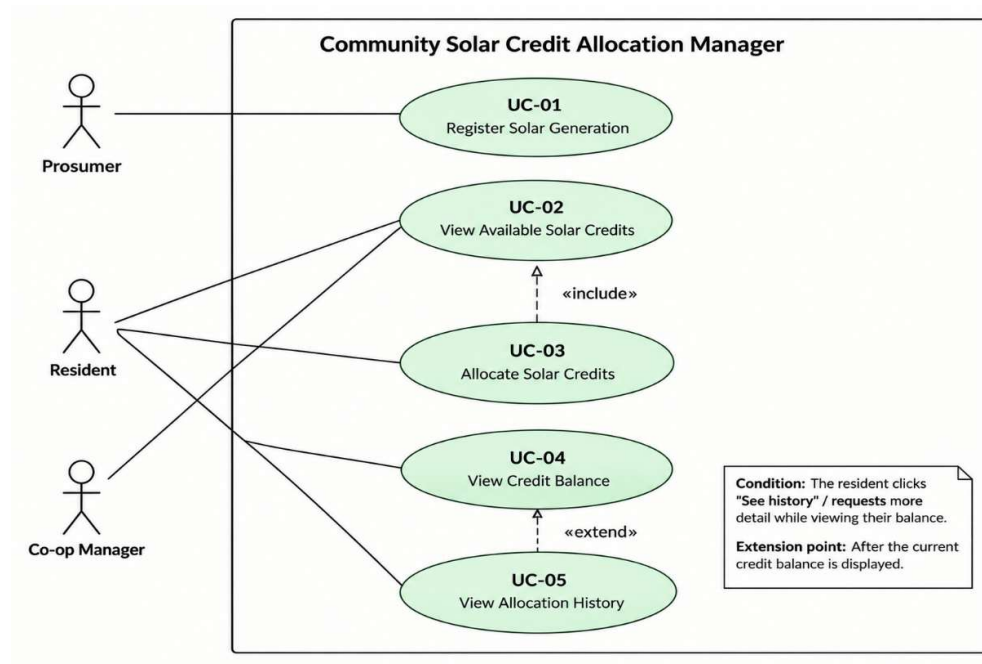
Actors

1. **Prosumer** – Generates and contributes solar energy to the community.
2. **Resident** – Receives and uses allocated solar credits.
3. **Co-op Manager** – Manages and allocates solar credits among residents.

Use Cases

1. UC-01 – Register Solar Generation
2. UC-02 – View Available Solar Credits
3. UC-03 – Allocate Solar Credits
4. UC-04 – View Credit Balance
5. UC-05 – View Allocation History
6. UC-06 – Receive Allocation Notification

UML Use-Case Diagram



Use-Case Flow

Use Case: UC-03 – Allocate Solar Credits

Main Success Scenario

1. The Co-op Manager selects **Allocate Solar Credits**.
2. The system displays the available solar credits.
3. The Co-op Manager selects the eligible Resident(s).
4. The Co-op Manager enters the amount of solar credits to be allocated.
5. The system validates the entered allocation.
6. The system updates the selected Resident's solar credit balance.
7. The system records the allocation in the allocation history.
8. The system displays a confirmation message indicating that the allocation was successful.
9. The use case ends successfully.

Alternate Flow – Insufficient Solar Credits

4a. Insufficient Credits

1. The system detects that the requested allocation is greater than the available solar credits.
2. The system displays an **“Insufficient Solar Credits”** message.
3. The Co-op Manager is asked to enter a lower allocation amount.
4. The Co-op Manager enters a revised allocation.
5. The system validates the revised allocation.
6. If the allocation is valid, the system continues from Step 6 of the Main Success Scenario.

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